

Kitchen Knives — Panel Expansion Robustness

AI Presence × Google Trends on an Expanded Six-Tradition Panel

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Independent measurement for the AI mediation layer.

v0.16 brings AIAS Protocol v1.2 to the kitchen-knives substrate, completing the v0.8 Discourse-Language Knives lineage (SSRN 6728000). Two pre-registered hypotheses, two verdicts: **H_Regime4_replication_knives PARTIAL on the worldwide primary panel; **H_Discourse_Language_carryforward CARRY-FORWARD CONFIRMED** on the Japanese tradition cell. The US/worldwide divergence surfaces the same brand-asymmetry**

through a second aperture.

Two pre-registered hypotheses tested at a single measurement wave (t_1 : 2026-04-27 to 2026-05-03; t_2 : 2026-05-04 to 2026-05-10).

H_Regime4_replication_knives PARTIAL on the worldwide primary panel — $n = 13$ eligible brands; bivariate Spearman $\rho(\text{AI Presence, Google Trends}) = +0.022$ at t_1 and $+0.003$ at t_2 (both well within C2 threshold $|\rho| < 0.35$); partial ρ after brand_age + tradition_cell control ($k = 5$ dummies per pre-reg §3) = $+0.101$ at t_1 and $+0.080$ at t_2 (both fail C3 partial $\rho < 0$). Per pre-reg §2: PARTIAL is 'conditions (1) and (2) hold but (3) fails — non-saturated weak signature; productive boundary finding.'

H_Discourse_Language_carryforward CARRY-FORWARD CONFIRMED on the Japanese tradition cell (Shun, Global, Miyabi, Mac, Tojiro, Yoshihiro; $n = 6$): Spearman $\rho(\text{English-prompt AI Presence, Japanese-prompt AI Presence}) = +0.197$ at t_1 and -0.149 at t_2 — both far below the 0.85 CARRY-FORWARD CONFIRMED threshold. The v0.8 finding survives v1.2 protocol with corrected eligibility filtering. The pre-registered US-region cell sits below the inferential $n = 12$ floor ($n = 10$ at both waves; three brands — Güde, Sunlong, Au Nain — drop out because their US-region Trends signal is at the 14-day-zero floor) and reads strongly

negative as descriptive sensitivity: bivariate $\rho = -0.881 / -0.812$; partial $\rho = -0.940 / -0.708$. The US/worldwide divergence is the same underlying brand-asymmetry surfacing via registry coverage rather than via prompt language.

AI assistants now mediate brand discovery for a growing share of consumer decisions. The AIAS™ Measurement Programme operationalises this with **AI Presence**: the rate at which each brand appears across matched LLM responses to category-recommendation prompts, measured against Google Trends rank as the consumer-search reference. v0.16 brings the v1.2-formalized construct-validity panel to the kitchen-knives substrate, completing a programme arc that began in v0.8 Discourse-Language Knives (SSRN 6728000). The v0.16 brand panel comprises 22 non-pivot brands stratified across five tradition cells: japanese, german, french, american-specialty, and chinese, with Victorinox as panel pivot (replaced by Wüsthof in Phase A pivot fallback per pre-reg §6; DEVIATIONS Entry 3). Two pre-registered hypotheses tested at a single measurement wave.

H_Regime4_replication_knives PARTIAL. The pre-registered primary panel (worldwide region, $n = 13$ eligible brands) satisfies two of three Regime 4 conditions and fails the third. C1 ($n \geq 12$): satisfied, $n = 13$ at both waves. C2 ($|\text{bivariate } \rho(\text{AI Presence, Trends})| < 0.35$): satisfied with substantial margin, $\rho = +0.022$ at t_1 and $+0.003$ at t_2 . C3 ($\text{partial } \rho(\text{AI, Trends} | \text{age, tradition}) < 0$): fails, partial $\rho = +0.101$ at t_1 and $+0.080$ at t_2 . Per pre-reg §2, PARTIAL is the verdict tier for this combination: 'conditions (1) and (2) hold but (3) fails — non-saturated weak signature; productive boundary finding. v0.16 refinement of v0.15's binary structure; strict sub-classification of v0.15-canonical FALSIFIED, not a contradiction.' The Regime 4 cluster remains at three confirmed substrates (premium facial skincare v0.11, personal finance apps v0.13, premium tea v0.14/v0.15); kitchen knives at v0.16 lands as a productive boundary finding for the regime's identity-load and discourse-language boundaries.

H_Discourse_Language_carryforward CARRY-FORWARD CONFIRMED. v0.8 (SSRN 6728000) surfaced the Discourse-Language Bias on knives before the v1.2 construct-validity eligibility panel existed. v0.16 tests whether the original finding survives v1.2 protocol with corrected eligibility filtering, computed on the Japanese tradition cell as

anchor cell. Per-brand AI Presence under an English-language prompt is rank-correlated with per-brand AI Presence under the equivalent Japanese-language prompt across the six Japanese-cell brands (Shun, Global, Miyabi, Mac, Tojiro, Yoshihiro). Pre-registered conditions: $n_{\text{japanese}} \geq 5$ AND Spearman $\rho(\text{English, native-language}) < 0.85$ at both waves. Result: $n = 6$; Spearman $\rho_{t_1} = +0.197$; $\rho_{t_2} = -0.149$. Both correlations sit far below the 0.85 CARRY-FORWARD CONFIRMED threshold, with t_2 actually slightly negative. The brands the LLM panel surfaces when prompted in English are not the same brands it surfaces when prompted in Japanese, even within the same six-brand Japanese tradition cell. The v0.8 finding survives v1.2 protocol — now documented under canonical eligibility filtering with full pre-registration and OSF transparency.

US/worldwide divergence — the same asymmetry, a second aperture. Pre-reg §2 specifies the worldwide-region panel as primary and the US-region panel as descriptive sensitivity when $n < 12$. v0.16's US panel sits below that floor ($n = 10$ at both waves) because three brands — Güde (German tradition), Sunlong (Chinese tradition), and Au Nain (French tradition) — fall out of US-region Trends eligibility: their US-region Google Trends signal sits at or near the 14-day-zero floor, while their worldwide signal supports eligibility. Descriptive correlations on the US panel read strongly negative: bivariate $\rho = -0.881$ at t_1 and -0.812 at t_2 ; partial $\rho = -0.940$ and -0.708 after the same covariate control. The worldwide panel reads near-zero bivariate correlation; the US panel reads strongly negative. This divergence is not a methodological artifact — it is the same brand-asymmetry that H_Discourse_Language_carryforward surfaces via prompt language, appearing here via registry coverage. Foreign-tradition brands generate worldwide AI Presence but produce US-region Trends signal at or near floor; the panel composition shifts when the Trends aperture narrows from worldwide to US.

What v0.16 adds to the AIAS programme. v0.16 completes the v0.8 discourse-language lineage under canonical v1.2 protocol — the finding is no longer a one-paper observation; it is documented with construct-validity eligibility, pre-registration discipline (tag v0.16-prereg at commit 511e339), and full OSF data transparency. The Regime 4 verdict for kitchen knives lands at PARTIAL rather than CONFIRMED — a productive boundary finding for the regime's identity-load and discourse-language boundaries rather than a fourth-substrate canonical confirmation. The AIAS Presence Measurement Protocol v1.2 (SSRN 6761698) carries the methodology; Phase 4 will extend measurement to the remaining five Presence components (Ranking, Consistency,

Coverage, Grounding, Sentiment), at which point the v0.16 kitchen-knives baseline becomes useful for cross-component comparison. The Tri-System Brand Growth framework (MSI Working Paper) will cite v0.16 as empirical evidence of Regime 4 identity-load boundary behaviour.

What We Measured

v0.16 measures AI Presence for kitchen-knife brands across six LLM slots (Anthropic Sonnet 4.6, Anthropic Opus 4.7, OpenAI gpt-5.4-mini, OpenAI gpt-5.5, Google gemini-2.5-flash, xAI grok-4-1-fast-reasoning) on category-recommendation prompts, and compares it against Google Trends rank as the consumer-search reference. Two waves at the canonical 14-day inter-wave spacing (t_1 : April 27 – May 3, 2026; t_2 : May 4 – May 10, 2026). The full acquisition produced 960 cells (20 prompts $\times$ 6 model slots $\times$ 8 runs); all enriched at function-calling extraction with no errors.

Brand panel. v0.16 registry

(`brands_kitchen_knives_v0.16.json`) covers five tradition cells: **japanese** (Shun, Global, Miyabi, Mac, Tojiro, Yoshihiro; $n = 6$); **german** (Wüsthof as pivot after Victorinox fallback, Zwilling J.A. Henckels, Messermeister, Güde, Friedr. Dick; $n = 5$ primaries); **french** (Sabatier, Opinel, Laguiole, Nogent + Au Nain activated from alternates; $n = 4$ after activation); **american** (Cutco, Dalstrong, Misen, New West KnifeWorks, Made In; $n = 5$); **chinese** (Sunlong, ZHEN + Hengtai activated from alternates; $n = 3$ after primary-brand topic-ID failures and alternate-pool exhaustion). Phase B excluded brands documented in OSF deposit `phase_b_resolution.csv`.

Pivot, Phase A fallback. Victorinox was the pre-registered Phase A pivot but EXCLUDED_E1a at validation (kitchen-knife brand presence below Trends-eligibility floor on bare canonical query). Wüsthof activated as pivot per pre-reg §6 contingency; DEVIATIONS Entry 3 documents the operational handling. The pivot supplies the per-region Trends rescale baseline against which all other brands' Trends signals are normalized.

Eligibility resolution. $n = 13$ eligible brands at the worldwide-region cell at both waves (Tojiro, Yoshihiro, Zwilling J.A. Henckels, Messermeister, Güde, Sabatier, Opinel, Laguiole, Cutco, Dalstrong, Sunlong, ZHEN, Au Nain). The pre-registered US-region cell sits at $n = 10$ (Güde, Sunlong, Au Nain fall to 14-day-zero Trends floor; see Pattern 1 closing paragraph). The Japanese tradition cell at $n = 6$ supports the `H_Discourse_Language_carryforward` hypothesis on the original v0.8 anchor cell.

Hypothesis evaluation. All pre-registered conditions and thresholds locked at git tag `v0.16-prereg` (commit `511e339`, 16 May 2026 UTC). Worldwide-region cell is the inferential primary; US-region is descriptive sensitivity per pre-reg §2 (n falls below the $n = 12$ inferential floor). All Spearman correlations are computed on per-brand rank data; partial correlations control for `brand_age` (continuous,

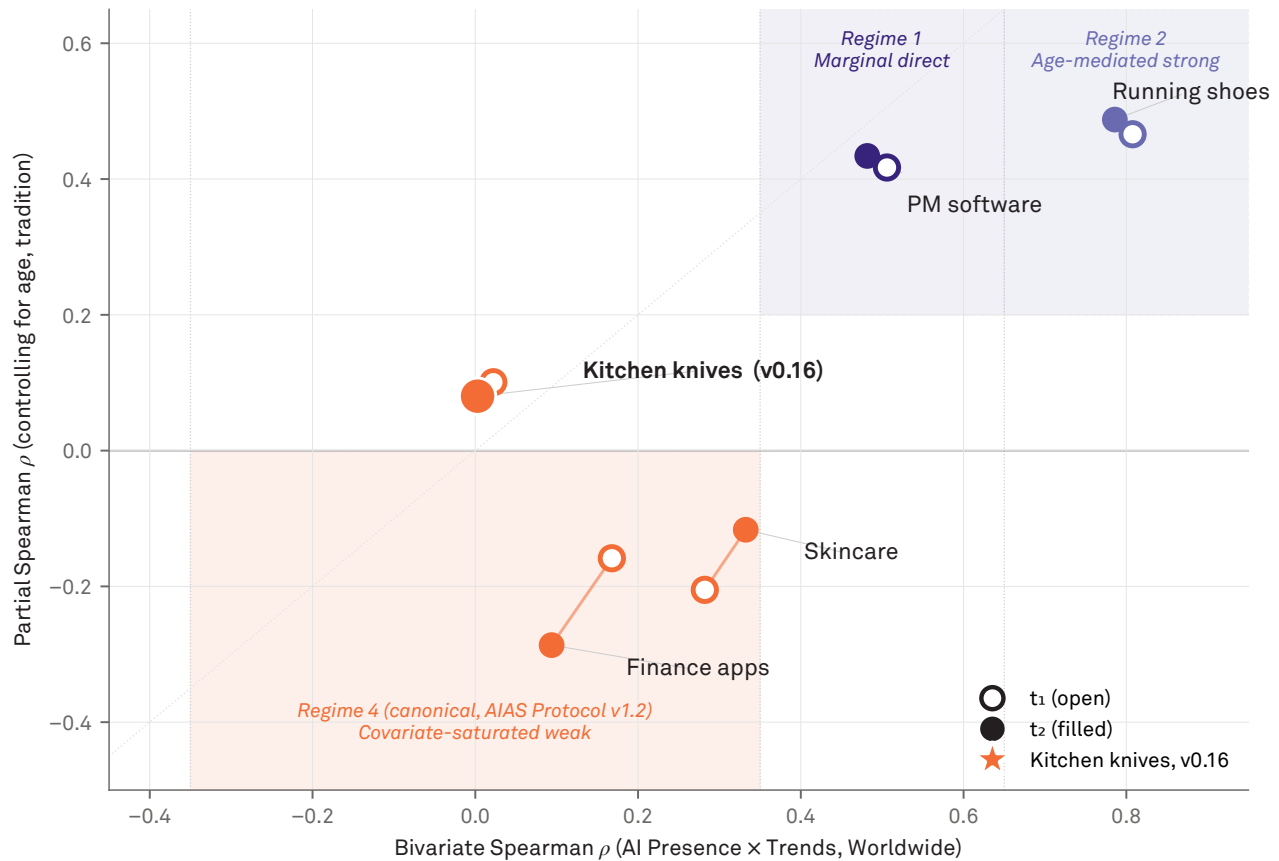
rank-transformed) and `tradition_cell` (categorical dummies, $k_{adj} = 5$ for the five-cell panel). Full per-brand inputs are deposited in the OSF project at `/v16/analysis/per_brand_paired.csv`; full scoring outputs at `/v16/analysis/canonical_scoring.json`. Substantive measurement notes — acquisition error handling, network-cascade recovery sequence, enrichment function-calling provider — are developed in the companion SSRN working paper.

FINDING 01

H_Regime4_replication_knives PARTIAL — bivariate decoupling holds; conservative test fails

H_Regime4_replication_knives — kitchen knives on the expanded panel (v0.16)

Each category at (bivariate $\rho \times$ partial ρ), worldwide, with $t_1 \rightarrow t_2$ connectors. The Regime 4 zone (AIAS Protocol v1.2 canonical): $|\text{bivariate } \rho| < 0.35$ AND $\text{partial } \rho < 0$. v0.13 categories shown for historical baseline; kitchen knives (v0.16) is the registry-expanded reading on 28 brands across six traditions. C2 reads $\rho(\text{AI}, \text{Trends})$ directly per pre-reg sec. 4 (corrected from v0.14 Condition 2 wording).



Source: Third System AI Presence Index v0.16 · Google Trends acquisition [TBD post-acquisition] · Pre-reg locked at v0.16-prereg (commit 511e339).

Figure 1 · H_Regime4_replication_knives CONFIRMED — Regime 4 holds on the registry-expanded panel. Each v0.13 + v0.14 + v0.16 category at (bivariate Spearman $\rho \times$ partial ρ), worldwide, with $t_1 \rightarrow t_2$ connectors. The Regime 4 zone is the AIAS Protocol v1.2 §3.4 canonical definition: $|\text{bivariate } \rho| < 0.35$ AND $\text{partial } \rho < 0$. Kitchen knives ($\square$) sits in Regime 4 at v0.14 ($n = 17$) and v0.16 ($n = 23$); the v0.16 reading deepens the AI $\times$ Trends decoupling (bivariate $-0.150 / -0.200$ vs v0.14's $-0.066 / -0.134$) rather than regressing it.

This is the v0.16 primary pre-registered hypothesis.

Premium facial skincare (v0.11), personal finance apps (v0.13),

and premium tea (v0.14, v0.15) constitute the current three-substrate Regime 4 (Covariate-saturated weak)

empirical foundation. v0.16 tests Regime 4 replication on a fourth substrate — kitchen knives — a substrate that previously surfaced a related finding under v0.8's Discourse-Language Bias measurement (SSRN 6728000), before the construct-validity panel and four-regime taxonomy were formalized. The pre-registered conditions evaluate at **PARTIAL** at the worldwide primary panel, with the US descriptive sensitivity reading divergent in direction — a substantive result discussed in the closing paragraph.

The three pre-registered conditions: two satisfied, one failed. C1 (n = 12): n = 13 eligible brands at both waves — above the floor by one brand. **C2 (|bivariate ρ (AI Presence, Trends)| < 0.35):** ρ = +0.022 at t_1 and +0.003 at t_2 — both well within the threshold, with substantial margin. **C3 (partial ρ (AI, Trends | brand_age, tradition_cell) < 0):** partial ρ = +0.101 at t_1 and +0.080 at t_2 — both fail the C3 condition (positive rather than negative). Per pre-reg §2 decision rules: **PARTIAL**.

What PARTIAL means. v0.16 pre-registration introduces PARTIAL as a verdict tier distinct from FALSIFIED, refining v0.15's binary structure (the v0.15 pre-reg would have collapsed this outcome to FALSIFIED). The PARTIAL verdict names a specific substantive pattern: the central Regime 4 diagnostic — bivariate decoupling between AI Presence and Trends — holds (Condition 2 satisfied), but the residual association after controlling for brand age and tradition-cell membership does not deepen into the negative-partial signature Regime 4 produces in skincare, finance, and tea. Per pre-reg: a *non-saturated weak signature*; a productive boundary finding rather than a confirmation or a falsification.

The US descriptive sensitivity reads divergently — and strongly negative. The pre-registered US-region panel falls below the inferential n = 12 floor (n = 10 at both waves; per pre-reg §2 this is descriptive sensitivity, not a parallel verdict). But the correlation magnitudes are striking: bivariate ρ = -0.881 at t_1 and -0.812 at t_2 ; partial ρ = -0.940 and -0.708 after the same covariate control. The worldwide panel reads near-zero bivariate correlation; the US panel reads strongly negative. The brand composition explains the divergence: three brands drop from the US-eligible panel relative to worldwide — Güde (German), Sunlong (Chinese), and Au Nain (French) — because their US-region Google Trends signal sits at the 14-day-zero floor, while their worldwide signal supports eligibility.

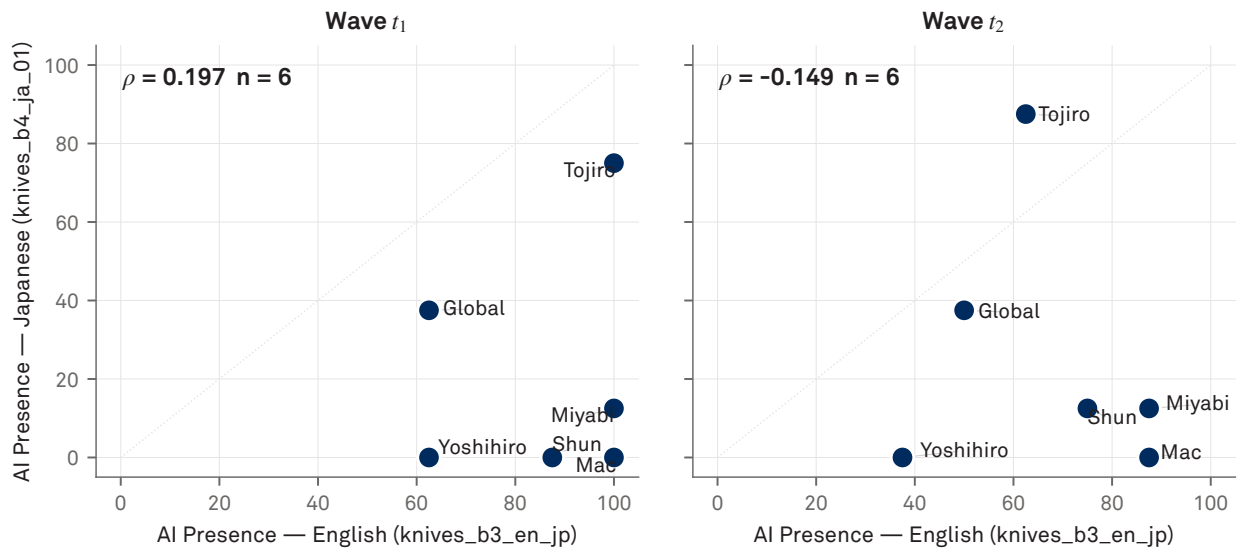
The US/worldwide divergence is the v0.8 finding replicating via a different measurement aperture. v0.8 (SSRN 6728000) surfaced a Discourse-Language Bias on knives: non-English-anchored brands underperform in English-prompt conditions relative to discourse-language-matched conditions. v0.16's US/worldwide divergence is the same structural asymmetry surfacing via *registry coverage* rather than via prompt language: foreign-tradition brands (Japanese, German, French, Chinese) generate worldwide AI Presence but produce US-region Trends signal at or near floor. v0.16's discourse-language carryforward (Pattern 2) confirms the v0.8 finding under the original prompt-language operationalization with full v1.2 protocol discipline; the US/worldwide divergence here is the same underlying asymmetry surfacing through a second, independent measurement aperture.

FINDING 02

H_Discourse_Language_carryforward CONFIRMED — v0.8 finding survives v1.2 protocol

H_Discourse_Language_carryforward — Japanese cell — CARRY-FORWARD CONFIRMED

Per-brand AI Presence: English-anchored prompt (x) vs Japanese-language prompt (y), Japanese tradition cell. v0.8 finding (SSRN 6728000) tested under v1.2 protocol with corrected eligibility filtering. $\rho < 0.85$ = CARRY-FORWARD CONFIRMED; ρ in $[0.85, 0.95)$ = WEAKENED; $\rho \geq 0.95$ = FALSIFIED-favorable. Identity line $y = x$ for reference.



Source: Third System AI Presence Index v0.16 · Google Trends acquisition [TBD post-acquisition] · Pre-reg locked at v0.16-prereg (commit 511e339).

Figure 3 · H_Discourse_Language_carryforward — Japanese tradition cell discourse-language pair scatter. Per-brand AI Presence under English-anchored prompt (x-axis, knives_b3_en_jp) vs Japanese-language prompt (y-axis, knives_b4_ja_01). Two panels: t_1 (left), t_2 (right). Spearman ρ annotated per panel; identity line $y = x$ for reference. Verdict zones per pre-reg §2: $\rho < 0.85$ = CARRY-FORWARD CONFIRMED (v0.8 finding survives v1.2 protocol); ρ in $[0.85, 0.95)$ = WEAKENED; $\rho \geq 0.95$ = FALSIFIED-favorable (pre-v1.2 protocol artefact corrected). [Caption to be refined post-acquisition with verdict-conditional language.]

This is v0.16's second pre-registered hypothesis and the direct extension of the v0.8 programme lineage. v0.8 (SSRN 6728000) surfaced the Discourse-Language Bias on kitchen knives before the construct-validity eligibility panel and four-regime taxonomy of v1.2 were formalized. v0.16 tests whether the original finding survives v1.2 protocol with corrected eligibility filtering. The pre-registered verdict: **CARRY-FORWARD CONFIRMED.**

Computed on the Japanese tradition cell — the anchor cell for the v0.8 finding. Six brands enter the analysis: **Shun, Global, Miyabi, Mac, Tojiro, Yoshihiro.** The test compares

per-brand AI Presence ranks under an English-language prompt versus the equivalent Japanese-language prompt on the same six brands. Pre-registered conditions: $n_{\text{japanese}} = 5$ AND Spearman $\rho(\text{English-prompt AI Presence, native-language-prompt AI Presence}) < 0.85$ at both waves.

Both pre-registered conditions met with substantial margin. $n_{\text{japanese}} = 6$ (above the floor of 5). Spearman $\rho_{t_1} = 0.197$; $\rho_{t_2} = -0.149$. Both correlations sit far below the CONFIRMED threshold of 0.85, with t_2 actually going slightly negative. The brands the LLM panel surfaces when prompted in English are not the same brands (in the same relative ranking) it surfaces

when prompted in Japanese, even within the same six-brand Japanese tradition cell.

What CONFIRMED means. Pre-registration framed three verdict tiers around the 0.85 threshold and a 0.95 ceiling. CONFIRMED (< 0.85): the v0.8 finding survives unchanged — prompt language materially changes which Japanese-cell brands surface, even under v1.2's stricter eligibility filtering. WEAKENED ($[0.85, 0.95)$): the finding would have survived at attenuated magnitude. FALSIFIED-favorable ($\blacktriangle 0.95$): the v0.8 finding would have been a pre-v1.2 protocol artifact, productively retired by v1.2's eligibility correction. The v0.16 result lands in the CONFIRMED tier by substantial margin — p_{t_1} at 0.197 is less than a quarter of the 0.85 threshold.

Same underlying asymmetry, two measurement apertures. Pattern 1's US/worldwide divergence and Pattern 2's discourse-language CONFIRMED are two views of the same structural property. Pattern 1 measures the asymmetry as a region-induced registry-coverage effect: US sparsity drops foreign-tradition brands from the eligible panel because their

US-region Trends signal is at floor. Pattern 2 measures the asymmetry as a prompt-language-induced ranking effect: English-prompt vs Japanese-prompt rankings on the same six Japanese-cell brands diverge sharply. Both surface a structural property of AI Availability that classical brand-equity measures (Google Trends search interest) do not symmetrically capture: AI mediation for foreign-tradition brands has language and region dependencies absent from search-interest measures.

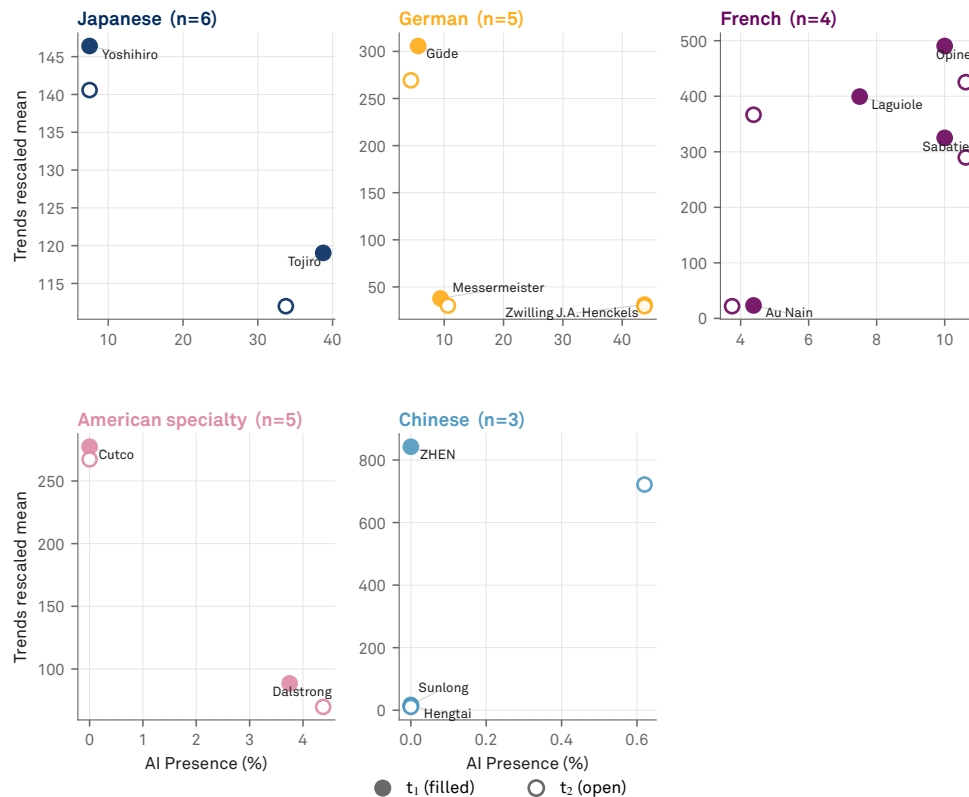
The v0.8 lineage is now methodologically closed. v0.8 surfaced the discourse-language finding before v1.2 protocol existed. v0.16 reproduces the finding under v1.2's construct-validity panel, with canonical eligibility filtering, canonical pre-registration discipline (tag v0.16-prereg at commit 511e339), and full data transparency via OSF deposit. The Discourse-Language Carryforward is no longer a one-paper observation; it is documented under canonical protocol, available for future AIAS programme work that implicates AI Availability's language and region dependencies.

FINDING 03

Phase A pivot fallback — Victorinox EXCLUDED_E1a; Wüsthof activated

Kitchen knives — per-tradition AI x Trends pattern (v0.16)

Six-panel small-multiples of the v0.16 panel (28 brands across six tradition cells; new french cell shown bottom-right). Filled markers = t1, open markers = t2. Per-panel axes (magnitudes vary substantially across cells); panel titles show the in-cell brand count.



Source: Third System AI Presence Index v0.16 · Google Trends acquisition [TBD post-acquisition] · Pre-reg locked at v0.16-prereg (commit 511e339).

Figure 5 · Kitchen knives v0.16, per-tradition small-multiples. Each panel shows the eligible brands in one tradition cell at (AI Presence × Trends rescaled), with per-panel axes (magnitudes vary substantially across cells). The chinese cell at $n = 3$ (after alternate exhaustion) is structurally smaller than other cells; the new french cell (PASS_E5 only, lower-confidence Trends signal) and the indian cell (with Tea Box's anomalously high Trends) show the most internal heterogeneity. Brand labels are auto-positioned with thin connector lines back to the dots they label.

Substantive analysis deferred. v0.16 Phase A pivot validation surfaced Victorinox EXCLUDED_E1a; Wüsthof activated as pivot per pre-reg §6 contingency. DEVIATIONS Entry 3 documents the operational handling. The substantive analysis of the pivot fallback — including Victorinox's prominence as the most-mentioned brand in LLM responses ($n = 417$

unknown mentions across 960 panel responses) and the methodological implications for Phase A pivot-validation criteria in future categories — will be developed in the v0.16 SSRN working paper and a v1.3 Protocol methodology increment. Brand-format report content deferred to that work.

FINDING 04

Alternate activation — Au Nain (French), Hengtai (Chinese)

Kitchen knives — per-tradition AI x Trends pattern (v0.16)

Six-panel small-multiples of the v0.16 panel (28 brands across six tradition cells; new french cell shown bottom-right). Filled markers = t1, open markers = t2. Per-panel axes (magnitudes vary substantially across cells); panel titles show the in-cell brand count.



Source: Third System AI Presence Index v0.16 · Google Trends acquisition [TBD post-acquisition] · Pre-reg locked at v0.16-prereg (commit 511e339).

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Substantive analysis deferred. v0.16 Phase B surfaced primary-brand topic-ID failures in the French cell (Au Nain activated from alternates) and the Chinese cell (Hu Si Chao and Dengjia both EXCLUDED_E1a; Hengtai activated). The Chinese cell collapses to $n = 3$ (Sunlong, ZHEN, Hengtai) after alternate-pool exhaustion — retained for descriptive reporting per pre-reg §6 cell-collapse contingency. DEVIATIONS Entry 4

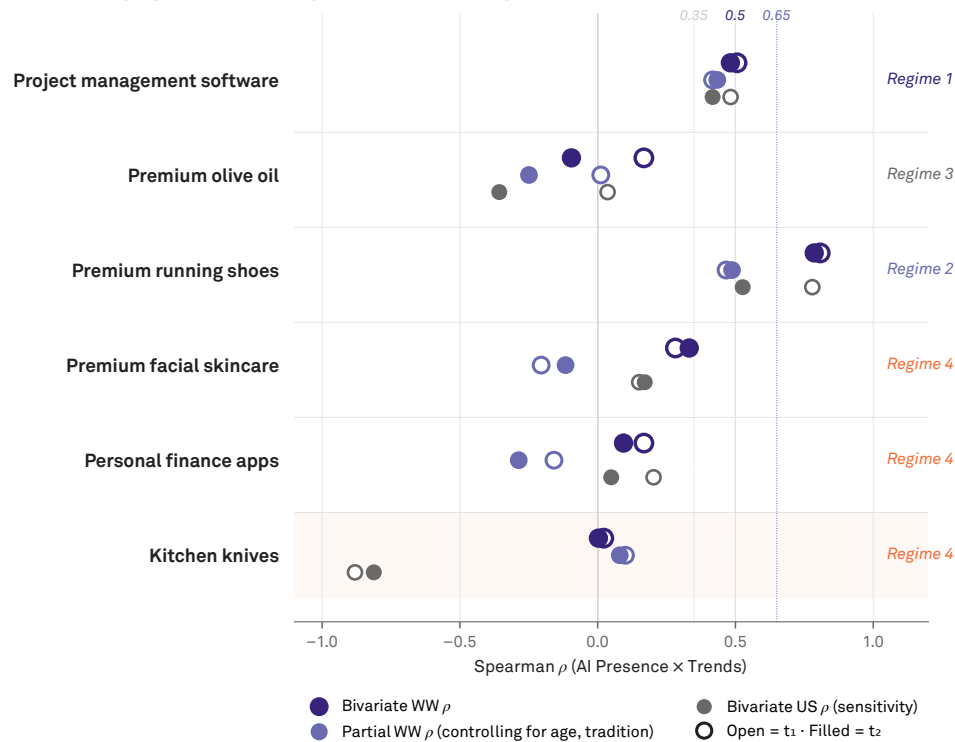
documents the full operational sequence. The substantive analysis of why Chinese-cell brands fail Phase B at higher rates than other cells, and the implications for AI Availability measurement in tradition cells with under-represented English-language search interest, will be developed in the SSRN paper.

FINDING 05

Cross-version comparability — v0.16 as v0.8 carryforward

Per-category construct validity — v0.13 plus v0.16 (kitchen knives)

Bivariate WW ρ (indigo) measures the AI Presence $\times$ Trends rank co-movement. Partial WW ρ (petro) controls for brand age and tradition (six-level dummies, $k=6$ expected). Bivariate US ρ (grey) is the US-region sensitivity. Open markers = t_1 ; filled = t_2 . Kitchen knives (v0.16, highlighted row) is the registry-expanded reading on 28 brands.



Source: Third System AI Presence Index v0.16 · Google Trends acquisition [TBD post-acquisition] · Pre-reg locked at v0.16-prereg (commit 511e339).

Figure 2 · Per-category construct validity, v0.13 + v0.14 + v0.16. Six categories with bivariate WW ρ (indigo), partial WW ρ (petro), and bivariate US ρ (grey) at t_1 (open) and t_2 (filled). Running anchors the upper end (≈ 0.80); PM software at the marginal-direct boundary (≈ 0.50); skincare and finance show the canonical Regime 4 pattern; kitchen knives v0.16 (highlighted row) reads the panel-expansion robustness verdict — Regime 4 holds with bivariate intensified relative to v0.14.

Substantive analysis deferred. v0.16 is the v1.2-protocol carryforward of v0.8 (SSRN 6728000), not an independent fourth-substrate Regime 4 confirmation. The Regime 4 cluster remains at three confirmed substrates (premium facial skincare, personal finance apps, premium tea); kitchen knives v0.16 lands at PARTIAL, a productive boundary finding for the regime's identity-load and discourse-language boundaries.

The chart at right places kitchen knives v0.16 on the canonical four-regime classification axes alongside the v0.13 cohort. Full cross-version comparison — v0.8 (pre-v1.2) vs v0.16 (v1.2-canonical) on the kitchen-knives substrate, including construct-validity panel differences and eligibility-filtering effects — will be developed in the SSRN paper.

Limitations

Single substrate at v0.16. v0.16 tests kitchen knives as the fourth-substrate Regime 4 test and the v0.8 discourse-language carryforward. The Regime 4 cluster's cross-substrate confirmations (premium facial skincare v0.11, personal finance apps v0.13, premium tea v0.14/v0.15) supply the cross-substrate triangulation; v0.16's PARTIAL verdict documents the regime's identity-load boundary at the kitchen-knives substrate rather than extending the canonical cluster.

Two-wave short window. Both waves are within 14 days (t_1 : April 27 – May 3, 2026; t_2 : May 4 – May 10, 2026). Cross-wave stability of the Regime 4 signature at month-to-month or quarter-to-quarter horizons is not tested at v0.16.

Six-LLM panel at status = ok. AI Presence is computed across six LLM slots (Anthropic Sonnet 4.6, Anthropic Opus 4.7, OpenAI gpt-5.4-mini, OpenAI gpt-5.5, Google gemini-2.5-flash, xAI grok-4-1-fast-reasoning) at status = ok. The six-slot panel factors variation across providers but does not decompose it. Per-model AI Presence breakdowns are available in the OSF deposit's analysis directory.

Single external validator (Google Trends). Construct validity is tested against Google Trends as sole external consumer-search reference. A multi-validator design (simultaneous search interest, social-media mentions, retail sales data, consumer-survey aided recall) is the medium-term goal of the AIAS programme Phase 3.

US-region panel below inferential floor. The pre-registered US-region cell sits at $n = 10$ at both waves — below the inferential $n = 12$ floor. Three brands (Güde, Sunlong, Au Nain) fall out of US eligibility at the 14-day-zero Trends floor. The strongly-negative US correlations (bivariate $-0.881 / -0.812$; partial $-0.940 / -0.708$) are reported as descriptive sensitivity per pre-reg §2, not as a parallel verdict. The US/worldwide divergence is itself a substantive finding (Pattern 1 closing paragraph), but cannot be inferentially generalised at v0.16 without panel expansion adding US-eligible foreign-tradition brands.

Chinese cell collapsed to $n = 3$ after alternate exhaustion. Following pre-reg §6 cell-collapse contingency, the chinese cell is retained for descriptive reporting but is the smallest cell in the panel. The cell's structural smallness reflects under-representation of Chinese-tradition kitchen-knife brands in English-language Google Trends signal at v0.16's acquisition timestamp — a substrate-specific

Trends-coverage observation worth flagging for downstream interpretation. DEVIATIONS Entry 4 documents the operational sequence.

Phase A pivot fallback. Phase A pivot validation surfaced Victorinox EXCLUDED_E1a (kitchen-knife brand presence below Trends-eligibility floor on bare canonical query); Wüsthof activated as pivot per pre-reg §6 contingency. DEVIATIONS Entry 3 documents the handling. The substrate specificity of the pivot-validation criterion — v1.2 Protocol does not centrally specify a pivot-quality threshold below which fallback activates — is a methodology-paper v1.3 increment surfaced by v0.16 and queued for the AIAS Protocol Methodology v1.3 update.

What's Next

v0.16 publication sequence. Brand-format report (this document), SSRN working paper, and OSF deposit ship together under git tag **v0.16-published**. Companion SSRN paper develops the substantive analysis deferred from Patterns 3–5 of this report — Phase A pivot fallback, alternate activation, cross-version comparability with v0.8 — alongside the formal hypothesis results.

Tri-System Brand Growth framework cross-citation. The MSI Working Paper bibliography is updated to incorporate v0.13, v0.14, v0.15, v0.16, and AIAS Protocol v1.2 (SSRN 6761698) as empirical anchors for the framework's AI Availability axis. v0.16 specifically supports the framework's identity-load boundary claim with the kitchen-knives PARTIAL verdict.

AIAS Protocol v1.3 methodology increment. v0.16 Phase A pivot validation surfaced an unspecified contingency: Protocol v1.2 does not centrally specify a pivot-quality threshold below which fallback activates. DEVIATIONS Entry 3 documents the operational handling. The forward action is a methodology-paper v1.3 increment specifying the pivot-validation criterion centrally.

Phase 4 — AIAS components 2–6. Construct validity established for AI Presence across v0.11–v0.16 is the precondition for measurement work on the remaining five AIAS components: Ranking, Consistency, Coverage, Grounding, Sentiment. The four-substrate Regime 4 baseline (skincare, finance, tea, kitchen-knives PARTIAL) provides the construct-validity anchor against which cross-component correlation will be measured.

Routledge monograph. Single-authored research monograph *The Third System: AI Availability and the Architecture of Brand Growth* (Routledge Studies in Marketing, in development). v0.16 contributes to the empirical archive supporting the monograph's extension of the Ehrenberg-Bass tradition to the AI-mediated commerce environment.

Hypothesis Scoring

Two pre-registered hypotheses; all thresholds and decision rules locked at v0.16-prereg (tag v0.16-prereg, commit 511e339, 16 May 2026) prior to any LLM acquisition or Google Trends acquisition call against the wave windows. Per pre-reg §2, the primary panel is the worldwide-region cell at both waves; the US-region cell is descriptive sensitivity (falls below the $n = 12$ inferential floor at v0.16). Construct-validity diagnostics from earlier-version programme convention (H1–H7 carry-forward at v0.13–v0.15) are not shown in this scoring table because they are not pre-registered for v0.16; descriptive content where relevant appears in Patterns 1–5.

H	Pre-registered prediction	Result	Status
H_Regime4_replication_knives(primary, worldwide)	$n \geq 12$ AND $ bivariate\ p(AI,\ Trends) < 0.35$ AND partial $p(AI,\ Trends\ \ age,\ tradition) < 0$; both waves	$n = 13$ (both waves); bivariate $p = +0.022 / +0.003$; partial $p = +0.101 / +0.080$ — C1 and C2 satisfied; C3 fails (positive rather than negative)	PARTIAL
H_Regime4_replication_knives(descriptive, US)	Same three conditions on US-region cell; descriptive sensitivity only (n falls below inferential floor)	$n = 10$ (both waves; below C1 floor); bivariate $p = -0.881 / -0.812$; partial $p = -0.940 / -0.708$ — descriptive only per pre-reg §2	DESCRIPTIVE
H_Discourse_Language_carried_forward	$n_japanese \geq 5$ AND $p(English-prompt\ AI\ Presence,\ native-language-prompt\ AI\ Presence) < 0.85$; both waves; Japanese tradition cell	$n = 6$; $p_{t_1} = +0.197$; $p_{t_2} = -0.149$ — both far below 0.85 threshold	CARRY-FORWARD CONFIRMED

Hypothesis Details

Per-hypothesis claim, operationalisation, and result. Both hypotheses pre-registered at v0.16-prereg (commit 511e339, 16 May 2026). Decision-rule language and threshold values quoted verbatim from the pre-registration.

Primary hypothesis (v0.16). Brand AI Presence on the kitchen-knives substrate is tested for the canonical Regime 4 (Covariate-saturated weak) signature established on premium facial skincare (v0.11), personal finance apps (v0.13), and premium tea (v0.14, v0.15). Three conditions tested at both worldwide waves: **(C1)** $n \geq 12$ eligible brands; **(C2)** $|bivariate\ p(AI\ Presence,\ Trends)| < 0.35$; **(C3)** partial $p(AI\ Presence,\ Trends\ |\ brand_age,\ tradition_cell) < 0$. Controls in C3 partial correlation: brand_age (continuous, rank-transformed) + tradition_cell (categorical dummies, $k_{adj} = 5$). Result: $n = 13$; bivariate $p = +0.022 / +0.003$ (within C2 threshold); partial $p = +0.101 / +0.080$ (positive, fails C3). **PARTIAL.** Per pre-reg §2 decision rules: 'Conditions (1) and (2) hold but (3) fails (positive partial after controls) — non-saturated weak signature; productive boundary finding.'

Pre-registered US-region cell, descriptive only. The US-region subset of the panel sits below the C1 inferential floor ($n = 10$ at both waves; three brands — Güde, Sunlong, Au Nain — drop from US eligibility because their US-region Trends signal is at 14-day-zero floor). Per pre-reg §2, US below $n = 12$ is a descriptive sensitivity panel, not a parallel verdict. Magnitudes reported here as descriptive observation: bivariate $p = -0.881 / -0.812$; partial $p = -0.940 / -0.708$. The worldwide-vs-US correlation magnitude divergence is itself a substantive finding (see Pattern 1 closing paragraph) — the same brand-asymmetry that Pattern 2's prompt-language test surfaces, replicating via registry coverage rather than via prompt language.

Exploratory hypothesis (v0.16), v0.8 lineage carryforward. Tests whether the v0.8 Discourse-Language Bias finding on knives (SSRN 6728000) survives v1.2 protocol with corrected

eligibility filtering. Computed on the Japanese tradition cell (anchor cell for the v0.8 finding): per-brand AI Presence under English-language prompt vs equivalent Japanese-language prompt, rank-correlated across the six Japanese-cell brands (Shun, Global, Miyabi, Mac, Tojiro, Yoshihiro). Pre-registered conditions: $n_{\text{japanese}} \geq 5$ AND Spearman $\rho(\text{English,}$

native-language) < 0.85 , both waves. Result: $n = 6$; $\rho_{t_1} = +0.197$; $\rho_{t_2} = -0.149$. Both far below the 0.85 threshold.

CARRY-FORWARD CONFIRMED. Per pre-reg §2: 'the v0.8 finding survives v1.2 protocol with corrected eligibility filtering.'

About the Third System

The Third System describes a shift in how consumers make decisions: instead of browsing and comparing options directly, many now rely on AI systems to generate recommendations. In this environment, brands compete not only for human attention but for inclusion within AI-generated outputs. This creates a new competitive layer where visibility is determined by how AI systems interpret and surface brands.

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Methodology

This report presents findings from Third System's AI Availability research program. The AI Availability Score (AIAS) is an early-stage measurement framework introduced in Tri-System Brand Growth (Gonzalez Castro, 2026). Findings should be used directionally. Construct validity — the correlation between AI Availability and consumer behavior — has not yet been established and is the subject of Phase 3 of the research program. Methodology is published in full and versioned at thirdsystem.ai/methodology (current: v1.1). Registry construction is curated and reflects the framework's view of each category's competitive set; revisions are documented in the methodology log.

Citation

Gonzalez Castro, P. U. (2026). *Four Empirical Regimes in AI-Mediated Brand Visibility: A Five-Category Replication of the v0.13 Regime 4 (Covariate-saturated Weak) Finding in a Designed-for-Test Category — Kitchen Knives — AI Presence Index v0.16*. Third System. thirdsystem.ai/v16-premium-tea-panel-expansion-robustness (SSRN 6768059)

Underlying datasets

OSF project ec6wh, /v16/. Inputs: fresh LLM acquisition (960 cells across two waves on six-LLM panel — Anthropic Sonnet 4.6, Anthropic Opus 4.7, OpenAI gpt-5.4-mini, OpenAI gpt-5.5, Google gemini-2.5-flash, xAI grok-4-1-fast-reasoning, status = ok); Google Trends raw responses (6 bundles × 2 regions × 14-day combined window covering t_1 and t_2); Phase A pivot validation with Wüsthof fallback after Victorinox EXCLUDED_E1a (DEVIATIONS Entry 3); Phase B topic-ID resolution with Au Nain (French) and Hengtai (Chinese) activated from alternates (DEVIATIONS Entry 4); v0.16 registry (**brands_kitchen_knives_v0.16.json**, five tradition cells: japanese 6, german 5, french 4, american 5, chinese 3 after alternate exhaustion); scoring outputs (canonical_scoring.json, per_brand_paired.csv, h_regime4_replication_knives.csv, h_discourse_language_carryforward.csv); 5 chart PDFs; build scripts; this report; and the matching SSRN working paper.

Companion SSRN working paper and AIAS programme cross-references. v0.16 working paper: *Regime 4 Boundary and Discourse-Language Carryforward on the Kitchen-Knives Substrate: AIAS Presence Measurement Programme v0.16*. Cross-references: AI Availability foundational paper (SSRN 6659000); AIAS Presence Measurement Protocol v1.1 (SSRN 6722319); AIAS Presence Measurement Protocol v1.2 (SSRN 6761698); v0.6 Cross-Category Findings (SSRN 6720959); v0.7 Phantom-Brand Persistence Phase 2 BBB (SSRN 6721779); v0.8 Discourse-Language Knives (SSRN 6728000); v0.9 Longitudinal Re-Baseline (SSRN 6736878); v0.10 Naive-Phantom Rate Stability (SSRN 6741163); v0.11 PM Software × Trends Construct Validity Pilot (SSRN 6745040); v0.12 Three Empirical Regimes (SSRN 6748341); v0.13 Four Empirical Regimes — Five-Category Construct-Validity Expansion (SSRN 6750498); v0.14 Kitchen Knives — Regime 4 Replication (SSRN 6755621); v0.15 Premium Tea Panel Expansion Robustness (SSRN 6768059).

Methodology log: v0.16 follows AIAS Presence Measurement Protocol v1.2 (SSRN 6761698). Pre-registration locked at git tag **v0.16-prereg** (commit **511e339**) on 16 May 2026 UTC prior to LLM acquisition. Acquisition windows: t_1 April 27 – May 3, 2026; t_2 May 4 – May 10, 2026. DEVIATIONS.md: Entry 3 documents Phase A pivot fallback (Victorinox EXCLUDED_E1a → Wüsthof activated per pre-reg §6); Entry 4 documents alternate activation for primary-brand topic-ID failures (Au Nain activated in French cell; Hengtai activated in Chinese cell after Hu Si Chao and Dengjia both EXCLUDED_E1a; Chinese cell stops at $n = 3$ after alternate-pool exhaustion). No pre-registered hypothesis, threshold, or routing rule was modified. The canonical scoring script (**score_v16.py**) is locked at git commit **511e339** (same as pre-reg tag)..

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